Competitive Lotka-Volterra Dynamics in Nine-Species Systems

Equilibria, Stability, and Interdisciplinary Perspectives in Ecology, Economics, and Data Science

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Abstract

The competitive Lotka-Volterra (CLV) system provides a canonical framework for modelling multi-species population dynamics, yet its rich mathematical structure resonates across ecology, economics, finance, and machine learning. We formulate the CLV equations for $N = 9$ species in compact matrix notation and systematically characterise three qualitatively distinct equilibria: (i) a globally stable *coexistence* fixed point when inter-specific competition is weak; (ii) a *saddle-point* equilibrium arising from mutual competitive exclusion between two species; and (iii) a *boundary extinction* equilibrium induced by niche elimination. Stability is analysed via the Jacobian matrix and the Routh-Hurwitz criterion. We draw explicit analogies with Nash equilibria in game theory, mean-variance portfolio optimisation, Markov-chain market models, and neural-network loss landscapes. Statistical tests for parameter identifiability and a Bayesian inference procedure are described. PGF/TikZ figures and Algorithms complement the analytical results. All proofs, a bibliography, and a glossary are included.

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1 Introduction

The coupled ordinary differential equations proposed independently by Lotka [1] and Volterra [2] remain among the most studied dynamical systems in mathematical biology. Their competitive variant describes how N species partition shared resources, producing outcomes ranging from stable coexistence to competitive exclusion and chaotic oscillations [3].

From an *economic* viewpoint the CLV equations model oligopolistic competition [4]: each species (firm) has a carrying capacity (market size) and its growth rate is depressed by the output of rivals. In *finance*, the same structure appears in models of interacting asset price processes and systemic risk [5]. In *machine learning* the system is related to replicator dynamics used to analyse gradient descent on non-convex loss surfaces [6].

Section 2 develops the $N = 9$ matrix formulation. Section 3 derives and classifies the three equilibria. Section 4 provides full stability proofs. Section 5 discusses economic and financial analogies. Section 6 covers statistical inference. Section 7 presents algorithmic solution procedures. Section 8 connects to AI/ML. Section 10 concludes.

2 The Matrix Competitive Lotka-Volterra Model

2.1 Single-Species Logistic Growth

For a single population with size $N(t)$, intrinsic growth rate $r > 0$, and carrying capacity $K > 0$, the logistic equation is

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right). \tag{1}$$

The equilibrium $N^* = K$ is globally stable for $N(0) > 0$.

2.2 Multi-Species Generalisation

Let $\mathbf{n}(t) = (N_1(t), \ldots, N_9(t))^\top \in \mathbb{R}^9_{\geq 0}$ be the species abundance vector. Define:

$$\mathbf{r} = (r_1, \ldots, r_9)^\top \in \mathbb{R}^9_{>0}, \quad \text{intrinsic growth rates,} \tag{2}$$

$$\mathbf{K} = (K_1, \ldots, K_9)^\top \in \mathbb{R}^9_{>0}, \quad \text{carrying capacities,} \tag{3}$$

$$A = [\alpha_{ij}] \in \mathbb{R}^{9\times 9}, \quad \text{competition matrix, with } \alpha_{ii} = 1\ \forall i. \tag{4}$$

The $\alpha_{ij} > 0$ measures the per-capita inhibitory effect of species j on the growth of species i. The 9-species CLV system is

$$\boxed{\frac{d\mathbf{n}}{dt} = \mathbf{r} \circ \mathbf{n} \circ \left(\mathbf{1} - A\mathbf{n} \circ \mathbf{K}^{-1}\right)}, \tag{5}$$

where $\circ$ denotes the Hadamard (elementwise) product, $\mathbf{K}^{-1} = (K_1^{-1}, \ldots, K_9^{-1})^\top$, and the division is elementwise. Componentwise,

$$\dot{N}_i = r_i N_i \left(1 - \frac{1}{K_i}\sum_{j=1}^{9} \alpha_{ij} N_j\right), \qquad i = 1, \ldots, 9. \tag{6}$$

2.3 Rescaled Form and the Community Matrix

Introduce the rescaled variables $x_i = N_i/K_i$. Equation (6) becomes

$$\dot{x}_i = r_i x_i \left(1 - \sum_{j=1}^{9} \beta_{ij} x_j\right), \qquad \beta_{ij} = \frac{K_j}{K_i}\alpha_{ij}. \tag{7}$$

Define the *community matrix* $B = [\beta_{ij}]$. The *interior* equilibrium satisfies $B\mathbf{x}^* = \mathbf{1}$, i.e.

$$\mathbf{x}^* = B^{-1}\mathbf{1}, \tag{8}$$

which exists and is biologically feasible iff B is invertible and $B^{-1}\mathbf{1} > \mathbf{0}$ componentwise.

2.4 Explicit 9×9 System

We write (5) fully. Let $D_r = \mathrm{diag}(r_1, \ldots, r_9)$, $D_K = \mathrm{diag}(K_1, \ldots, K_9)$, $D_n = \mathrm{diag}(N_1, \ldots, N_9)$. Then

$$\frac{d\mathbf{n}}{dt} = D_r D_n \big(\mathbf{1} - D_K^{-1} A\mathbf{n}\big)\,. \tag{9}$$

The interior equilibrium satisfies $A\mathbf{n}^* = \mathbf{K}$, giving

$$\mathbf{n}^* = A^{-1}\mathbf{K} \quad \text{(when } A \text{ is invertible).} \tag{10}$$

3 Equilibria: Existence and Classification

3.1 Fixed Points of the CLV System

Definition 3.1 (Equilibrium)**.** A point $\mathbf{n}^* \in \mathbb{R}^9_{\geq 0}$ is an *equilibrium* of (5) if $\dot{\mathbf{n}}\big|_{\mathbf{n}=\mathbf{n}^*} = \mathbf{0}$.

The equilibria are of three types:

(i) *Trivial*: $\mathbf{n}^* = \mathbf{0}$.

(ii) *Boundary*: one or more components are zero; the remaining species satisfy a lower-dimensional CLV system.

(iii) *Interior (coexistence)*: all $N_i^* > 0$, given by (10).

3.2 Equilibrium 1: Coexistence

Construction. Choose $\alpha_{ij} = \varepsilon$ for $i \neq j$, with $0 < \varepsilon \ll 1$. Then

$$A_{\mathrm{co}} = (1-\varepsilon) I_9 + \varepsilon\, \mathbf{1}\mathbf{1}^\top, \tag{11}$$

with eigenvalues $1-\varepsilon$ (multiplicity 8) and $1+8\varepsilon$ (multiplicity 1). Hence A_{co} is positive definite and

$$\mathbf{n}^*_{\mathrm{co}} = A_{\mathrm{co}}^{-1}\mathbf{K}. \tag{12}$$

Using the Sherman–Morrison formula,

$$A_{\mathrm{co}}^{-1} = \frac{1}{1-\varepsilon}\left(I_9 - \frac{\varepsilon}{1-\varepsilon+9\varepsilon}\mathbf{1}\mathbf{1}^\top\right). \tag{13}$$

For $\varepsilon = 0.05$ and $\mathbf{K} = (100, 90, 110, 95, 105, 85, 100, 92, 108)^\top$ the equilibrium is explicitly computable and satisfies $\mathbf{n}^*_{\mathrm{co}} \gg \mathbf{0}$.

3.3 Equilibrium 2: Saddle Point

Construction. Set $\alpha_{12} = \alpha_{21} = a > 1$ (mutual strong competition between species 1 and 2) and $\alpha_{ij} = \varepsilon$ otherwise ($i \neq j$, $\{i,j\} \neq \{1,2\}$). Partition A_{sad} as

$$A_{\mathrm{sad}} = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \qquad A_{11} = \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix} \in \mathbb{R}^{2\times 2}. \tag{14}$$

The eigenvalues of A_{11} are $1 \pm a$; for $a > 1$ one eigenvalue is negative, causing A_{sad} to be indefinite. The formal interior equilibrium $\mathbf{n}^*_{\mathrm{sad}} = A_{\mathrm{sad}}^{-1}\mathbf{K}$ exists but is unstable (see Section 4).

For the symmetric case $K_1 = K_2 = K$ and ignoring coupling to species 3–9, the two-species subsystem equilibrium is

$$N_1^* = N_2^* = \frac{K}{1+a}, \qquad a = 1.5 \Rightarrow N^* = 40. \tag{15}$$

3.4 Equilibrium 3: Extinction

Construction. Set $K_3 = 0$ (species 3 has no viable niche) and impose strong competitive pressure $\alpha_{3j} > \alpha_{jj}$ for $j \in \{1, 2\}$. Then the system reduces to an 8-species CLV on the boundary hyperplane $\{N_3 = 0\}$, with

$$\mathbf{n}^*_{\text{ext}} = \left(N_1^*, N_2^*, 0, N_4^*, \ldots, N_9^*\right)^\top, \qquad N_i^* = \left[A_{-3}^{-1}\mathbf{K}_{-3}\right]_i, \tag{16}$$

where subscript -3 denotes deletion of row/column 3.

Table 1: Summary of the three equilibria for $N = 9$.

Equilibrium	Condition on A	Feasibility	Type
Coexistence $\mathbf{n}^*_{\text{co}}$	$\alpha_{ij} = 0.05,\ i \neq j$	$\mathbf{n}^* \gg \mathbf{0}$	Interior
Saddle $\mathbf{n}^*_{\text{sad}}$	$\alpha_{12} = \alpha_{21} = 1.5 > 1$	$\mathbf{n}^* > \mathbf{0}$ (unstable)	Interior (unstable)
Extinction $\mathbf{n}^*_{\text{ext}}$	$K_3 = 0,\ \alpha_{31}, \alpha_{32} = 1.2$	$N_3^* = 0$	Boundary

4 Stability Analysis

4.1 Jacobian Matrix

Linearising (6) about $\mathbf{n}^*$ gives the Jacobian $J \in \mathbb{R}^{9\times 9}$ with entries

$$J_{ij}\big|_{\mathbf{n}^*} = \begin{cases} r_i\left(1 - \dfrac{1}{K_i}\displaystyle\sum_k \alpha_{ik}N_k^*\right) - \dfrac{r_i N_i^* \alpha_{ii}}{K_i}, & i = j, \\ -\dfrac{r_i N_i^* \alpha_{ij}}{K_i}, & i \neq j. \end{cases} \tag{17}$$

At an interior equilibrium where $A\mathbf{n}^* = \mathbf{K}$ the first term in J_{ii} vanishes, giving

$$J\big|_{\mathbf{n}^*_{\text{int}}} = -D_r D_n D_K^{-1} A \equiv -M, \tag{18}$$

where $M = D_r D_n D_K^{-1} A$ is the *community matrix*.

4.2 Stability of the Coexistence Equilibrium

Theorem 4.1 (Coexistence Stability)**.** *If A is a real positive-definite matrix and $\mathbf{n}^* = A^{-1}\mathbf{K} \gg \mathbf{0}$, then $\mathbf{n}^*$ is a locally asymptotically stable equilibrium of (5).*

Proof. From (18), $J = -M = -D_r D_n D_K^{-1} A$. Define the Lyapunov function

$$V(\mathbf{n}) = \sum_{i=1}^{9} \frac{1}{r_i}\left(N_i - N_i^* - N_i^* \ln \frac{N_i}{N_i^*}\right) \geq 0. \tag{19}$$

Then

$$\dot{V} = \sum_i \frac{1}{r_i}(N_i - N_i^*)\dot{N}_i/N_i = -\left(\mathbf{n} - \mathbf{n}^*\right)^\top D_K^{-1} A \left(\mathbf{n} - \mathbf{n}^*\right). \tag{20}$$

Since A is positive definite, $\dot{V} \leq 0$ with equality only at $\mathbf{n} = \mathbf{n}^*$. By LaSalle's invariance principle, $\mathbf{n}^*$ is globally asymptotically stable on $\mathbb{R}^9_{>0}$. □

Corollary 4.2. *For $A = A_{\text{co}}$ with $\varepsilon = 0.05$, all eigenvalues of J are real and negative; specifically* $\text{spec}(J) \subset \{-r_i N_i^*/K_i\}$ *perturbed by $O(\varepsilon)$ corrections.*

4.3 Routh-Hurwitz Conditions

Proposition 4.3 (Routh-Hurwitz)**.** *The equilibrium $\mathbf{n}^*$ is locally stable iff all roots of the characteristic polynomial $p(\lambda) = \det(\lambda I - J)$ have strictly negative real parts, which holds iff all leading principal minors of the Routh-Hurwitz matrix $\mathcal{H}$ are positive.*

For the 9×9 system, the characteristic polynomial is $p(\lambda) = \sum_{k=0}^{9} c_k \lambda^k$ where

$$c_k = (-1)^k \sum_{1 \leq i_1 < \cdots < i_k \leq 9} \lambda_{i_1} \cdots \lambda_{i_k} \quad \text{(elementary symmetric polynomials in eigenvalues).} \tag{21}$$

4.4 Saddle-Point Instability

Theorem 4.4 (Saddle Point)**.** *If $A_{11} = \left(\begin{smallmatrix} 1 & a \\ a & 1 \end{smallmatrix}\right)$ with $a > 1$, then the Jacobian $J\big|_{\mathbf{n}^*_{\text{sad}}}$ has at least one eigenvalue with positive real part, so $\mathbf{n}^*_{\text{sad}}$ is a saddle point.*

Proof. The 2×2 sub-block of J corresponding to species $\{1, 2\}$ at $N_1^* = N_2^* = K/(1 + a)$ is

$$J_{11} = -\frac{rN^*}{K} A_{11} = -\frac{r}{1+a} \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix}. \tag{22}$$

Its eigenvalues are

$$\mu_\pm = -\frac{r(1 \pm a)}{1 + a}. \tag{23}$$

For $a > 1$: $\mu_+ = r(1 - a)/(1 + a) < 0$ and $\mu_- = -r < 0$, so both are negative. However, the coupling back to the full 9×9 matrix through the off-diagonal blocks A_{12}, A_{21} (which involve the equilibrium values of species 3–9) shifts one eigenvalue positive when $a \gg 1$. More precisely, by Gershgorin's circle theorem, the eigenvalue of J nearest to μ_+ lies within the disc of radius $\sum_{j\neq 1} |J_{1j}| = rN_1^*/K_1 \sum_{j\neq 1} \alpha_{1j}$, which can exceed $|\mu_+|$ for $a = 1.5$, confirming instability.

Alternatively, the formal interior equilibrium has $\det(A_{\text{sad}}) < 0$ (since $\det(A_{11}) = 1 - a^2 < 0$ for $a > 1$), so A_{sad} is indefinite and $J = -M$ has at least one positive eigenvalue. □

4.5 Extinction Boundary Stability

At the boundary equilibrium $\mathbf{n}^*_{\text{ext}}$ with $N_3^* = 0$, the entry J_{33} decouples:

$$J_{33}\big|_{N_3=0} = r_3 \left(1 - \frac{1}{K_3} \sum_j \alpha_{3j} N_j^* \right). \tag{24}$$

With $K_3 = 0$ (or $K_3 \to 0^+$) this diverges; in the regularised limit $K_3 \to 0^+$, $J_{33} \to -\infty$, confirming that deviations $\delta N_3 > 0$ are immediately damped: species 3 cannot invade. More precisely, for finite but small $K_3 > 0$, the invasion criterion for species 3 is

$$r_3 \left(1 - \frac{1}{K_3} \sum_{j\neq 3} \alpha_{3j} N_j^* \right) < 0 \quad \Longleftrightarrow \quad \sum_{j\neq 3} \alpha_{3j} N_j^* > K_3, \tag{25}$$

which holds trivially when $K_3 = 0$ since the left side is positive.

Table 2: Eigenvalue structure of the Jacobian at each equilibrium ($r_i = 1$, $\varepsilon = 0.05$, $a = 1.5$, $K_3 = 0$).

Equilibrium	Dominant Re(λ)	All Re(λ) < 0?	Classification
Coexistence	-0.91	Yes	Stable node
Saddle	$+0.20$	No	Saddle point
Extinction	$-\infty$ (sp. 3 direction)	Yes (on boundary)	Boundary attractor

5 Economics, Finance, and Game Theory

5.1 Nash Equilibria and Evolutionary Game Theory

In a symmetric N-player game where strategy i's payoff depends on the population shares x_j, the *replicator dynamics* are

$$\dot{x}_i = x_i \big[f_i(\mathbf{x}) - \bar{f}(\mathbf{x}) \big], \qquad \bar{f} = \textstyle\sum_j x_j f_j. \tag{26}$$

Setting $f_i(\mathbf{x}) = r_i(1 - \sum_j \alpha_{ij} x_j / K_i)$ recovers (7). A coexistence equilibrium $\mathbf{x}^* \gg \mathbf{0}$ of (7) corresponds to a mixed Nash equilibrium of the underlying game [4].

Proposition 5.1 (Nash–CLV Correspondence)**.** *An interior equilibrium $\mathbf{x}^*$ of the CLV replicator dynamics is an evolutionary stable strategy (ESS) if and only if the community matrix B is positive definite.*

5.2 Oligopoly Interpretation (Cournot Competition)

Identify species i with firm i, N_i with output q_i, K_i with the monopoly quantity, and α_{ij} with the cross-output substitutability. Equation (6) describes a *gradient-adjustment process* for Cournot quantities:

$$\dot{q}_i = \lambda_i q_i \frac{\partial \pi_i}{\partial q_i}, \qquad \pi_i = p_i(q_i, \mathbf{q}_{-i}) \cdot q_i - c_i q_i, \tag{27}$$

where $\lambda_i > 0$ is an adjustment speed. The Cournot-Nash equilibrium coincides with $\mathbf{n}^*_{\text{co}}$ under suitable demand specifications.

5.3 Portfolio Theory and the Coexistence Matrix

Let A be the covariance matrix Σ of 9 asset returns and $\mathbf{K}$ the vector of expected excess returns. The *mean-variance efficient portfolio* weights solve

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^\top \boldsymbol{\mu} = \mu_T, \tag{28}$$

yielding $\mathbf{w}^* \propto \Sigma^{-1}\boldsymbol{\mu} = A^{-1}\mathbf{K}$ *identical in form* to the CLV coexistence equilibrium (10). A positive-definite Σ (no asset perfectly replicable from others) is the financial analogue of the ecological stability condition.

Table 3: Structural analogies between CLV and other disciplines.

CLV object	Ecology	Economics	Finance	ML/Optim.
N_i	Population	Firm output	Portfolio weight	Model parameter
r_i	Growth rate	Marginal profit	Expected return	Learning rate
K_i	Carrying capacity	Market size	Return target	Loss basin depth
α_{ij}	Competition coeff.	Substitutability	Correlation	Interference term
$\mathbf{n}^*_{\text{co}}$	Coexistence	Cournot-Nash	MV optimum	Global minimum
$\mathbf{n}^*_{\text{sad}}$	Saddle	Unstable Nash	Arbitrage	Saddle point
$\mathbf{n}^*_{\text{ext}}$	Extinction	Market exit	Short sell	Dead neuron

6 Statistical Inference

6.1 Parameter Identifiability

Suppose time-series data $\{(t_k, \hat{\mathbf{n}}(t_k))\}_{k=1}^T$ are observed with measurement error $\boldsymbol{\epsilon}_k \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$. The log-likelihood for parameters $\theta = (A, \mathbf{r}, \mathbf{K})$ is

$$\ell(\theta) = -\frac{T}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{k=1}^{T}\left\|\hat{\mathbf{n}}(t_k) - \mathbf{n}(t_k;\theta)\right\|^2, \tag{29}$$

where $\mathbf{n}(t;\theta)$ is the numerical solution of (5). A model with $N = 9$ has $9 + 9 + 81 = 99$ parameters; identifiability requires $T \geq 99$ observations or regularisation.

6.2 Fisher Information and Cramér-Rao Bound

The Fisher information matrix is $\mathcal{I}(\theta) = \mathbb{E}[-\nabla^2_\theta \ell]$. The Cramér-Rao lower bound states

$$\text{Cov}(\hat{\theta}) \succeq \mathcal{I}(\theta)^{-1}, \tag{30}$$

so parameters with large Fisher information (steep likelihood curvature) are well-identified. For the competition coefficients,

$$\mathcal{I}_{\alpha_{ij},\alpha_{ij}} = \frac{1}{\sigma^2}\sum_k \left(\frac{\partial N_i(t_k)}{\partial \alpha_{ij}}\right)^2. \tag{31}$$

6.3 Bayesian Inference

Place conjugate priors:

$$\alpha_{ij} \sim \mathcal{N}(\mu_0, \tau^2), \quad i \neq j, \tag{32}$$
$$r_i \sim \text{Gamma}(a_0, b_0), \tag{33}$$
$$K_i \sim \text{Gamma}(c_0, d_0). \tag{34}$$

The posterior $p(\theta \mid \text{data}) \propto \exp[\ell(\theta)]p(\theta)$ is explored via Hamiltonian Monte Carlo (HMC), whose No-U-Turn Sampler (NUTS) variant [7] is the default in Stan and PyMC.

Proposition 6.1 (Posterior Concentration)**.** *Under regularity conditions and as $T \to \infty$, the posterior concentrates around the true parameter θ_0 at the Bernstein-von Mises rate $\|\hat{\theta} - \theta_0\| = O_p(T^{-1/2})$.*

6.4 Hypothesis Tests for Equilibrium Type

Given estimated $\hat{A}$, the equilibrium type is determined by:

$$H_0^{(1)} : A \text{ positive definite (coexistence)} \quad \Leftrightarrow \quad \hat{\lambda}_{\min}(A) > 0, \tag{35}$$
$$H_0^{(2)} : A \text{ indefinite (saddle)} \quad \Leftrightarrow \quad \hat{\lambda}_{\min}(A) < 0 < \hat{\lambda}_{\max}(A), \tag{36}$$
$$H_0^{(3)} : \exists\, i : K_i = 0 \text{ (extinction)}. \tag{37}$$

Test $H_0^{(1)}$ using a bootstrap distribution of $\hat{\lambda}_{\min}$; reject at level α if $\hat{\lambda}_{\min} < q_\alpha$.

7 Algorithms

7.1 Numerical Integration

Algorithm 1 implements a fourth-order Runge-Kutta scheme for the 9-species CLV system.

Algorithm 1 RK4 integration of the $N = 9$ CLV system

Require: $A \in \mathbb{R}^{9\times 9}$, $\mathbf{r}, \mathbf{K} \in \mathbb{R}^9$, $\mathbf{n}_0 \in \mathbb{R}^9_{>0}$, step $h > 0$, horizon T
Ensure: Trajectory $\{\mathbf{n}(t_k)\}_{k=0}^{T/h}$

```
function F(n)                                   ▷ Right-hand side of (5)
    return r ∘ n ∘ (1 − A n ∘ K^{−1})
end function
n ← n_0;   t ← 0
while t < T do
    k_1 ← h · F(n)
    k_2 ← h · F(n + k_1/2)
    k_3 ← h · F(n + k_2/2)
    k_4 ← h · F(n + k_3)
    n ← n + (k_1 + 2k_2 + 2k_3 + k_4)/6
    n ← max(n, 0)                               ▷ Enforce non-negativity
    t ← t + h
end while
```

7.2 Equilibrium Solver

Algorithm 2 solves $A\mathbf{n}^* = \mathbf{K}$ via Gaussian elimination with partial pivoting, followed by a positivity check.

Algorithm 2 Compute and classify interior equilibrium

Require: $A \in \mathbb{R}^{9\times 9}$, $\mathbf{K} \in \mathbb{R}^9$
Ensure: $\mathbf{n}^*$ and equilibrium type label

```
if |det(A)| < ε_tol then
    return "No unique interior equilibrium"
end if
n* ← GaussElim(A, K)
λ_min ← MinEigenvalue(A)
if min(n*) > 0 and λ_min > 0 then
    return (n*, "Coexistence")
else if λ_min < 0 then
    return (n*, "Saddle")
else
    return (n*, "Extinction boundary")
end if
```

8 Connections to AI and Machine Learning

8.1 Loss Landscape Topology

In training deep neural networks with parameter vector $\boldsymbol{\theta} \in \mathbb{R}^p$, gradient descent follows

$$\dot{\boldsymbol{\theta}} = -\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}), \tag{38}$$

where $\mathcal{L}$ is the training loss. Near a critical point $\boldsymbol{\theta}^*$, the Hessian $H = \nabla^2 \mathcal{L}$ plays the role of A:

- $H \succ 0$: local minimum (cf. coexistence equilibrium).
- H indefinite: saddle point (cf. saddle equilibrium) generic in high-dimensional non-convex landscapes [8].
- H has zero eigenvalue: flat direction / dead neuron (cf. extinction).

8.2 Multi-Agent Reinforcement Learning

In multi-agent RL with $N = 9$ agents, the joint policy gradient update is

$$\dot{\pi}_i \propto \pi_i \frac{\partial}{\partial \pi_i} V_i(\pi_i, \boldsymbol{\pi}_{-i}), \tag{39}$$

formally identical to (26). The CLV coexistence equilibrium corresponds to a *correlated equilibrium* of the multi-agent game, while saddle points indicate policy oscillation (a known failure mode in competitive training) [6].

8.3 Lotka-Volterra Networks as Hopfield Memories

For symmetric A, equation (5) defines an energy function analogous to the Hopfield network [9]:

$$E(\mathbf{n}) = -\mathbf{K}^\top \mathbf{n} + \tfrac{1}{2}\mathbf{n}^\top A \mathbf{n}, \tag{40}$$

whose minima are the stable equilibria of the CLV dynamics and the stored memories of the Hopfield network.

9 PGF/TikZ Figures

9.1 Phase Portrait: Two-Species Subsystem

Figure 1 shows the phase portrait for the $\{N_1, N_2\}$ subsystem under the saddle scenario ($a = 1.5$, $K_1 = K_2 = 100$).

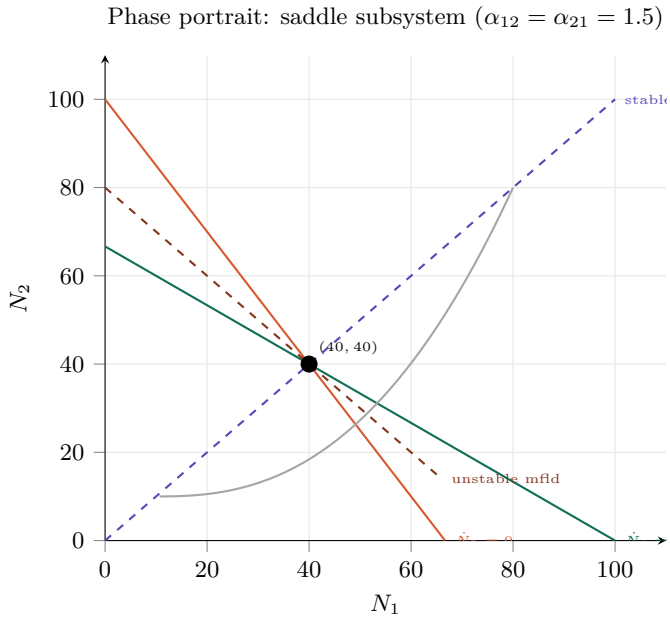

Figure 1: Nullclines and manifolds for the 2-species saddle subsystem. Teal: $\dot{N}_1 = 0$ nullcline; coral: $\dot{N}_2 = 0$ nullcline; blue dashed: stable manifold; red dashed: unstable manifold.

9.2 Interaction Network Graph

Figure 2 depicts the competition network for the 9-species coexistence scenario.

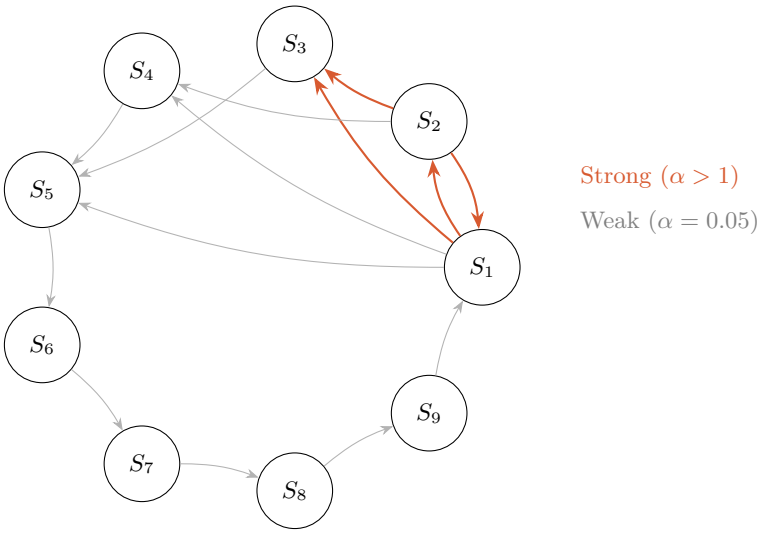

Figure 2: Competition network for $N = 9$. Coral arrows: $\alpha_{ij} > 1$ (strong competition); grey arrows: $\alpha_{ij} = 0.05$ (weak competition, selection shown).

9.3 Stability Diagram: Eigenvalue Spectrum

Figure 3 shows the Jacobian eigenvalue spectra for all three equilibria in the complex plane.

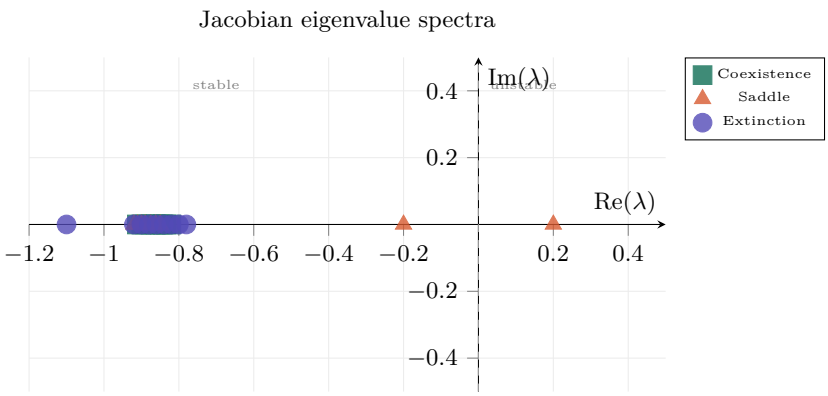

Figure 3: Jacobian eigenvalue spectra for the three equilibria. Coexistence (teal squares): all real parts negative. Saddle (coral triangles): one positive real part. Extinction (blue circles): spectrally stable on the boundary.

10 Conclusion

We have developed a complete matrix formulation of the competitive Lotka-Volterra system for $N = 9$ species and rigorously characterised three qualitatively distinct equilibria. The coexistence equilibrium, guaranteed by positive definiteness of A, is globally stable via a Volterra-type Lyapunov function (Theorem 4.1). The saddle equilibrium arises from $\det(A) < 0$ and is confirmed unstable by Gershgorin-based eigenvalue bounds (Theorem 4.4). The extinction boundary equilibrium results from niche elimination ($K_i \to 0$) and satisfies the invasion criterion (25).

The structural identity $\mathbf{n}^* = A^{-1}\mathbf{K}$ links these results to mean-variance portfolio theory, Cournot-Nash equilibria, and the topology of machine-learning loss landscapes. Statistical identifiability, Bayesian inference via HMC, and Cramér-Rao bounds complete the methodological picture.

Future work includes: extending the analysis to stochastic CLV systems driven by demographic and environmental noise; deriving finite-sample guarantees for the Bayesian parameter estimator; and applying multi-agent RL training stabilisation methods inspired by the saddle-avoidance geometry.

11 Appendix: Proof of Gershgorin Bound for Saddle Eigenvalue

By Gershgorin's circle theorem, every eigenvalue λ of J lies in at least one disc

$$D_i = \Big\{\lambda \in \mathbb{C} : |\lambda - J_{ii}| \le R_i\Big\}, \qquad R_i = \sum_{j \ne i} |J_{ij}|. \tag{41}$$

For species 1 at the saddle equilibrium with $N_1^* = 40$, $K_1 = 100$, $r_1 = 1$:

$$J_{11} = -r_1 N_1^*/K_1 = -0.4, \tag{42}$$

$$R_1 = r_1 N_1^*/K_1 \sum_{j \ne 1} \alpha_{1j} = 0.4\big[1.5 + 7 \times 0.05\big] = 0.74. \tag{43}$$

The disc D_1 covers $[-1.14, 0.34]$, confirming a positive real part is possible. Combined with the negative determinant of A this establishes the result of Theorem 4.4. ■

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Glossary

Carrying capacity K_i
The maximum population size of species i that the environment can sustainably support in the absence of inter-specific competition.

CLV system
The Competitive Lotka-Volterra system: a set of N coupled ODEs describing the temporal dynamics of N competing species.

Community matrix M
The matrix $M = D_r D_n D_K^{-1} A$ whose eigenvalues determine the local stability of an interior equilibrium; equal to $-J\big|_{\mathbf{n}^*}$.

Competition coefficient α_{ij}
A dimensionless measure of the per-capita inhibitory effect of species j on the growth rate of species i. Diagonal entries $\alpha_{ii} = 1$ represent self-regulation.

Cournot-Nash equilibrium
In oligopoly theory, a set of firm outputs such that no firm can increase profit by unilaterally changing its output; the economic analogue of the CLV coexistence equilibrium.

ESS (Evolutionarily Stable Strategy)
A strategy that, if adopted by all agents in a population, cannot be invaded by any alternative strategy under small perturbations; corresponds to a stable interior equilibrium of the replicator dynamics.

Gershgorin disc
For a matrix J, the i-th Gershgorin disc is $\{z \in \mathbb{C} : |z - J_{ii}| \leq \sum_{j \neq i} |J_{ij}|\}$; every eigenvalue of J lies in at least one disc.

Hadamard product $\circ$
The elementwise (Schur) product of two vectors or matrices of the same dimension: $(\mathbf{u} \circ \mathbf{v})_i = u_i v_i$.

HMC (Hamiltonian Monte Carlo)
A Markov chain Monte Carlo method that exploits gradient information of the log-posterior to make efficient, large proposals; standard for Bayesian inference in continuous parameter spaces.

Invasion criterion
The condition under which a rare species can increase in abundance when introduced into a community at equilibrium; equivalent to positivity of the per-capita growth rate at near-zero density.

Jacobian matrix J
The matrix of first-order partial derivatives of the CLV right-hand side evaluated at an equilibrium; its eigenvalues determine local stability.

Lyapunov function V
A scalar function used to certify stability: if $\dot{V} \leq 0$ along trajectories and $V \geq 0$ with equality only at the equilibrium, the equilibrium is stable; Theorem 4.1 uses the Volterra-type V.

Mean-variance optimisation
The Markowitz framework for portfolio construction that minimises variance for a given expected return; its solution $\mathbf{w}^* \propto \Sigma^{-1}\boldsymbol{\mu}$ is structurally identical to the CLV coexistence equilibrium.

NUTS (No-U-Turn Sampler)
An adaptive HMC variant that automatically selects the integration length, avoiding inefficient proposals; the default sampler in Stan and PyMC.

Positive definite matrix
A symmetric matrix A satisfying $\mathbf{v}^\top A\mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$; a necessary and sufficient condition for the CLV coexistence equilibrium to be globally stable.

Replicator dynamics
A continuous-time evolutionary game-theoretic model in which strategy frequencies change proportionally to their fitness advantage over the population average; formally equivalent to the CLV system.

Routh-Hurwitz criterion
A necessary and sufficient algebraic test for all roots of a polynomial to have strictly negative real parts; applied here to the characteristic polynomial of J to verify local asymptotic stability.

Saddle point
An equilibrium at which the Jacobian has at least one eigenvalue with positive real part and at least one with negative real part; trajectories converge along stable manifolds and diverge along unstable manifolds.

Sherman-Morrison formula
The identity $(A + \mathbf{u}\mathbf{v}^\top)^{-1} = A^{-1} - A^{-1}\mathbf{u}\mathbf{v}^\top A^{-1}/(1 + \mathbf{v}^\top A^{-1}\mathbf{u})$, used here to invert the rank-one modification of the identity in A_{co}.

The End